the fuel production should be allocated between main product and co -products. Within such conversion processes, emissions are allocated using their energy content, the so -called

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the fuel production should be allocated between main product and co -products. Within such conversion processes, emissions are allocated using their energy content, the so -called "energy allocation" approach. Where co -products allocation cannot be performed based on their energy content (e.g. Oxygen resulting from water electrolysis for H 2 production), other methods such as mass allocation, market revenue allocation (also known as "economic allocation"), could be considered on a case-by-case basis.

3.9 A co-product is defined as "an outcome of a production process, which has economic value and elastic supply (intended as the existence of a clear evidence of the causal link between feedstock market value and the quantity of feedstock that can be produced)".

3.10 This definition applies also when a raw material used to produce the fuels is a waste (no economic value) or a residue (unavoidably produced and with negligible economic value, needing further processing to be used in the main conversion process). In case the feedstock is a waste, a residue or a by -product, emissions considered as WtT start at the feedstock

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needing further processing to be used in the main conversion process). In case the feedstock is a waste, a residue or a by -product, emissions considered as WtT start at the feedstock collection point onwards until the point of use of the final fuel/energy product.

3.11 According to the IPCC Guidelines for National Greenhouse Gas Inventories ("the IPCC Guidelines"),4 any carbon in the fuel derived from biomass should be reported as an information item and not included in the sectoral or national totals to avoid double counting, since the net emissions from biomass are already accounted for in the Agriculture Forestry and Other Land Use (AFOLU) sector at a national level.

3.12 The scope of the IMO LCA guidelines does not affect or change the IPCC Guidelines. According to the IPCC Guidelines, international waterborne navigation (international bunkers) is grouped under "Mobile combustion" under the Energy sector, but emissions from fuel used by ships in international transport should not be included in national totals in national GHG inventories.

3.13 A fuel batch may be a mix of fuels made from various feedstocks and sources

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by ships in international transport should not be included in national totals in national GHG inventories.

3.13 A fuel batch may be a mix of fuels made from various feedstocks and sources (e.g. by blending 20% biodiesel into fossil MGO) and/or through different production pathways. The calculation should be done using the weighted averages of the energy of the various fuel components. Relevant information should accompany each component fuel in the FLL. Blended fuels should be included in the certification schemes and relevant GHG default or

actual emission factors (gCO2/MJ) determined in proportion to the energy of each fuel part of the blend.

4 WELL-TO-TANK (WtT)

4.1 The pathway of each relevant marine fuel should be clearly described and the GHG emissions during each step of the fuel pathway should be calculated. Specific GHG emissions of a specific non -conventional and non -fossil fuel's pathway may take into acco unt different characteristics across geographic regions, where feedstock production and/or conversion occurs, as appropriate.

4 2006 IPCC Guidelines for National Greenhouse Gas Inventories

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characteristics across geographic regions, where feedstock production and/or conversion occurs, as appropriate.

4 2006 IPCC Guidelines for National Greenhouse Gas Inventories

4.2 Any further reference in this document to a "fuel pathway" should be understood to include the feedstock structure (the so -called nature/carbon source and feedstock type pair) and the production or conversion process (noting that the same feedstock and fuel type pair can have a different production or conversion process).

4.3 The aim of the WtT methodology is to quantify and evaluate the GHG intensity of fuel production, including all steps mentioned in figure 2. The carbon feedstock and production pathway of a fuel should be identified in order to apply the methodology and is included as part of the FLL. The production steps to be included in the WtT are presented in figure 2.

Figure 2: Generic well-to-tank supply chain

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pathway of a fuel should be identified in order to apply the methodology and is included as part of the FLL. The production steps to be included in the WtT are presented in figure 2.

Figure 2: Generic well-to-tank supply chain

4.4 The WtT GHG emission factor (gCO 2eq/MJ(LCV) fuel or electricity) is calculated according to Equation (1).

Equation (1)

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Term	Units	Explanation
	gCO2eq / MJ(LCV)	Emissions associated with the feedstock extraction/cultivation/acquisition/recovery
	gCO2eq / MJ(LCV)	Emissions (annualized emissions (over 20 years) from carbon stock changes caused by direct land-use change)5
	gCO2eq / MJ(LCV)	Emissions associated with the feedstock processing and/or transformation at source and emissions associated with the conversion of the feedstock to the final fuel product, including electricity generation
	gCO2eq / MJ(LCV)	Emissions associated with the feedstock transport to conversion plant, and the emissions associated with the finished fuel transport and storage, local delivery, retail storage and bunkering

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	gCO2eq / MJ(LCV)	Emissions associated with the feedstock transport to conversion plant, and the emissions associated with the finished fuel transport and storage, local delivery, retail storage and bunkering
	gCO2eq / MJ(LCV)	Emissions (annualized emission savings (over 20 years) from soil carbon accumulation via improved agricultural management)6

5 Pending further methodological guidance to be developed by the Organization, the value of parameter should be set to zero.

6 Pending further methodological guidance to be developed by the Organization, the value of parameter should be set to zero.

Term	Units	Explanation
	gCO2eq / MJ(LCV)	Emissions credit from carbon capture and storage (e ccs), that have not already been accounted for in e p. This should properly account the avoided emissions through the capture and sequestration of emitted CO 2, related to the extraction, transport, processing and distribution of fuel (c sc). From the

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have not already been accounted for in e p. This should properly account the avoided emissions through the capture and sequestration of emitted CO 2, related to the extraction, transport, processing and distribution of fuel (c sc). From the above-mentioned emission credit, all the emissions resulting from the process of capturing (e cc) and transporting (e t) the CO2 up to the final storage (including the emissions related to the injection, etc.) need to be deducted.
This element should be calculated with the following formula:

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